The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 July 1999.

INCH-POUND

MIL-PRF-19500/429F
21 April 1999
SUPERSEDING
MIL-S-19500/429E
31 August 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY
TYPES 1N5615, 1N5617, 1N5619, 1N5621, 1N5623,
1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, 1N5623UL,
1N5615US, 1N5617US, 1N5619US, 1N5621US, 1N5623US,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, fast recovery power rectifier diodes. Four levels of product assurance are provided for each device as specified in MIL-PRF-19500. Two levels of product assurance are provided each unencapsulated device type.

1.2 Physical dimensions. See figures 1 (similar to D0-7), 2, 3, 4, and 5 (JANHC and JANKC).

1.3 Maximum ratings.

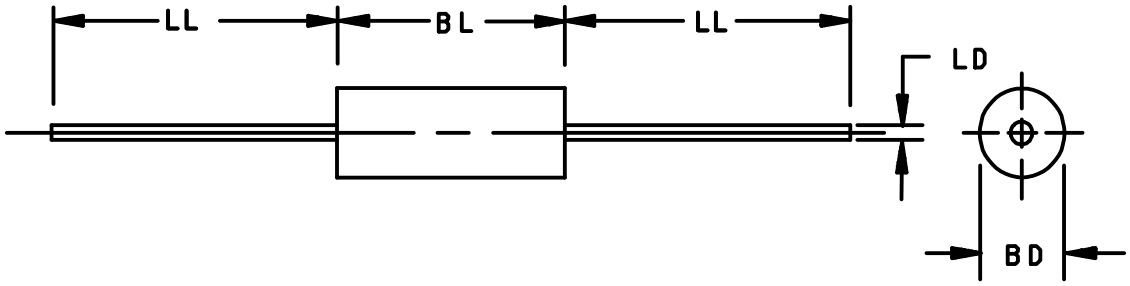
Types 5/	V_R	V_{RWM}	I_O		t_{rr}	T_{STG} and T_J	I_{FSM} T_A = +100°C I_O = 750 mA dc t_P = 8.3 ms	Barometric pressure (reduced)	$R_{\theta JL}$ 4/
			T_A = +55°C 1/ 2/	T_A = +100°C 2/ 3/					
	V dc	V(pk)	A dc	mA dc	ns	°C	A(pk)	mmHg	
1N5615	200	200	1	750	150	-65 to +175	25	N/A	See
1N5617	400	400	1	750	150	-65 to +175	25	8	Fig.
1N5619	600	600	1	750	250	-65 to +175	25	8	6
1N5621	800	800	1	750	300	-65 to +175	25	33	
1N5623	1,000	1,000	1	750	500	-65 to +175	25	33	

1/ From 1 A dc at T_A = +55°C, to .75 A dc at T_A = +100°C, derate linearly at 5.56 mA/°C.
2/ For the 1 A dc rating at +55°C and the 750 mA dc rating at +100°C, no special mounting, heat sinking, or forced-air flow across exposed areas of the device is required.
3/ From .75 A dc at T_A = +100°C, to 0 A dc at T_A = +175°C, derate linearly at 10 mA/°C.
4/ $R_{\theta JL} \leq 38$°C/W at L = .375 inch (9.53 mm); for UL and US suffix type devices. $R_{\theta JEC}$ = 7.0°C/W.
5/ Electrical characteristics for UL and US suffix are identical to the corresponding non suffix device.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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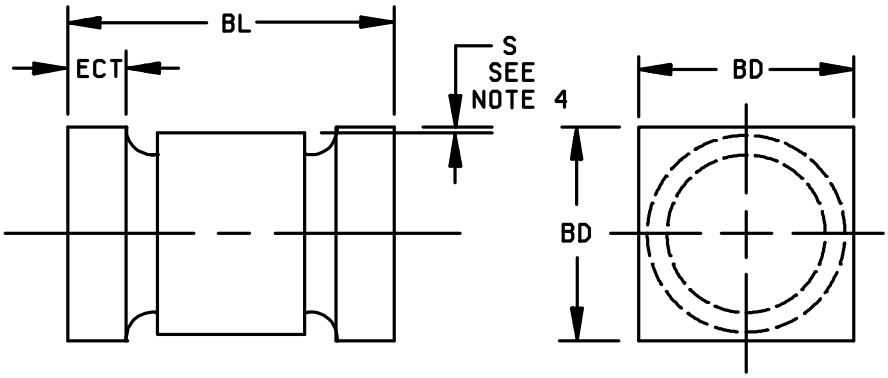


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.065	.110	1.65	2.79	3
LD	.026	.033	.66	.84	
BL	.130	.225	3.30	5.71	4
LL	1.00	1.30	25.40	33.02	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension BD shall be measured at the largest diameter.
4. Dimension BL shall include all uncontrolled areas of the device.

FIGURE 1. Physical dimensions (for non-UL and non-US suffix devices only).

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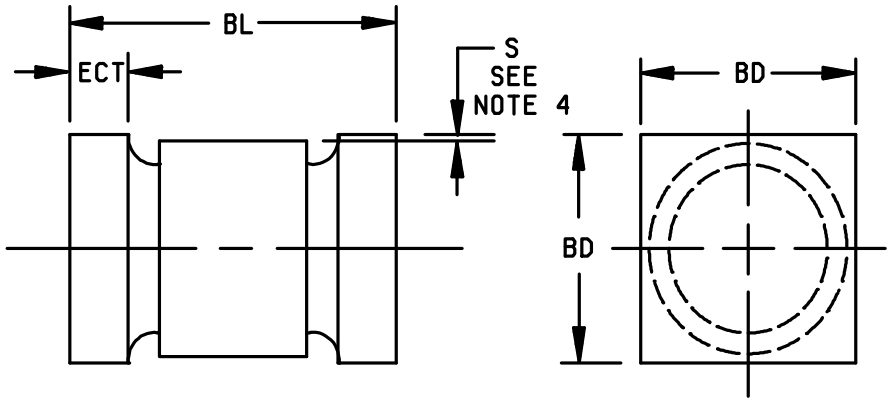


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.168	.225	4.27	5.72
ECT	.019	.028	0.48	0.71
S	.003	---	0.08	---
BD	.091	.125	2.31	3.18

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions for types 1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, and 1N5623UL, (surface mount devices) (D-5A).

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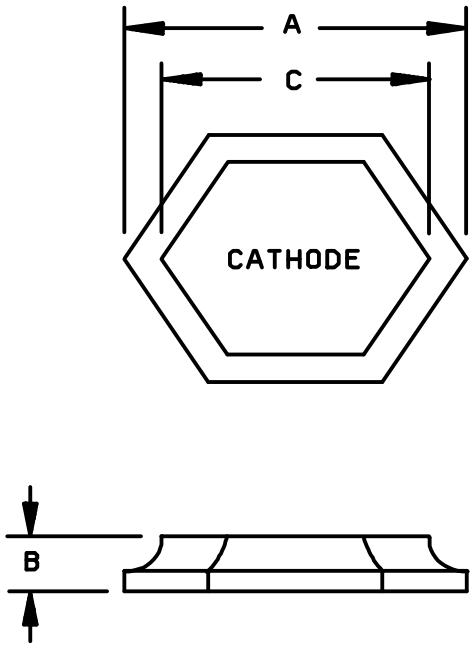


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.168	.200	4.27	5.08
ECT	.019	.028	0.48	0.71
S	.003	---	0.08	---
BD	.091	.103	2.31	2.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Physical dimensions for types 1N5615US, 1N5617US, 1N5619US, 1N5621US, and 1N5623US, (surface mount devices) (D-5A).

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A - version

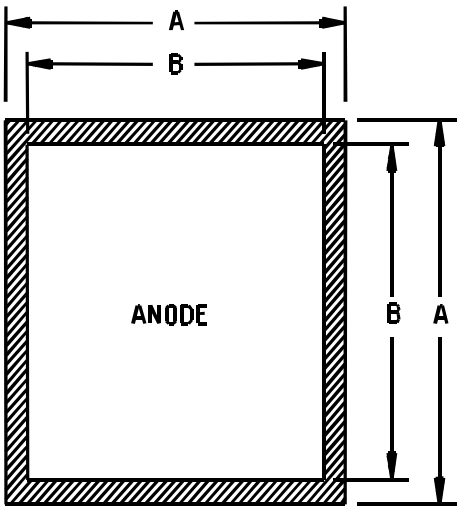
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.047	.053	1.19	1.35
B	.007	.011	0.18	0.28
C	.033	.037	0.84	0.94

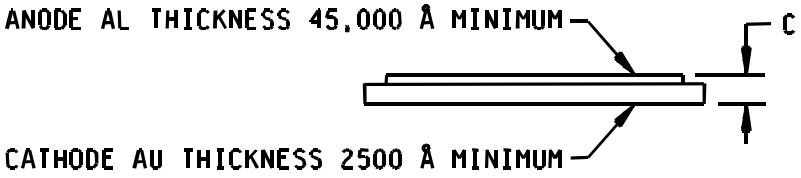
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:

Top metal: Gold 10,000 Å minimum.
Back metal: Gold 4,000 Å minimum.

FIGURE 4. Physical dimensions, JANHCA and JANKCA die.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	1.00	1.09
B	.021	.025	0.53	0.64
C	.008	.012	0.20	0.31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:

Top metal: Aluminum (anode)..... 45,000 Å minimum.
Back metal: Gold (cathode) ..... 2,500 Å minimum.

FIGURE 5. Physical dimensions, JANHCB and JANKCB die.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Method for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions shall be as specified in MIL-PRF-19500 and herein.

D.U.T - - - - Device under test.

UL - - - - Unleaded or surface mounted diodes with square end-caps.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1 (similar to D0-7), 2 (surface mount, UL suffix), 3 (surface mount, US suffix), and figure 4 and 5 (die) herein. Except for JANHC and JANKC, all devices shall be metallurgically bonded-thermally-matched-noncavity-double plug construction, utilizing a category I bond, in accordance with MIL-PRF-19500. The U version shall be structurally identical to the non-U version except for lead configuration.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, and as specified herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.4.1 Marking of UL and US version. For UL and US version only, all marking (see 3.5) may be omitted from the body, but shall be retained on the initial container.

3.5 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UL and US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end or a contrasting color band may be used.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.8 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

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4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 JANHC and JANKC devices. Qualification for die shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{R1} and V_F	Not applicable
11	I_{R1} and V_F, $\Delta I_{R1} \leq 100$ percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc	I_{R1} and V_F
12	See 4.3.1	See 4.3.1
13 2/	Subgroups 2 and 3 of table I herein: $\Delta I_{R1} \leq 100$ percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc. Scope display evaluation (see 4.5.4).	Subgroup 2 of table I herein: $\Delta I_{R1} \leq 100$ percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc. Scope display evaluation (see 4.5.4).

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.

2/ $Z_{\theta JX}$ need not be repeated at step 13 if performed earlier in the screening sequence.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038, condition B of MIL-STD-750, T_A = room ambient as designated in 4.5 of MIL-STD-750, V_R = full rated V_{RWM} (see 1.3); f = 50-60 Hz; I_O = 1 A dc, T_{EC} = +100°C to +125°C for surface mount devices (UL and US version).

4.3.2 Screening (JANHC or JANKC). Screening of die shall be in accordance with MIL-PRF-19500.

4.3.3 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101; maximum limit = 4.5°C/W.

4.3.3.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record value $Z_{\theta JX}$) and shall be supplied to the qualifying activity on a 500 piece sample from the qualification lot prior to qualification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$ in group A, subgroup 2 inspection:

a. I_M measuring current - - - - - - - 10 mA.

b. I_H forward heating current - - - - - 10 A.

c. t_H heating time - - - - - - - - - - 10 ms.

d. t_{MD} measurement delay time - - - - - 100 µs maximum.

The maximum limit for $Z_{\theta JX}$ in group A, subgroup 2 is $Z_{\theta JX}$ (maximum) = 4.5°C/W.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 and as specified in table II herein. $Z_{\theta JX}$ need not be performed. See subgroup conditions for delta limits when applicable. Leaded samples may be used in lieu of surface mount devices (UL and US version) for life tests.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	4066	I_{FSM} = 25 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, I_O = 750 mA dc; V_R = rated V_{RWM} (see 1.3); T_A = +100°C.
B4	1036	I_O = 1 A dc minimum; T_A = Room ambient as defined in MIL-STD-750; V_R = rated V_{RWM} (see 4.5.3); $t_{on} = t_{off}$ = 3 minutes minimum, 2,000 cycles; f = 50-60 Hz.
B5	1027	T_A = +150°C minimum, no heat sink, I_O = 1 A dc, minimum; adjust I_O and T_A as required to give an average T_J = +275°C +0°C, -35°C (see 4.5.3).
B6	3101 or 4081	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) ≤ 38°C/W, L = .375 inch (9.53 mm) (see figure 6). For surface mount devices (UL and US version) $R_{\theta JEC}$ ≤ 7°C/W.
B7	---	Peak reverse power, see 4.5.5 and figure 7 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; Sampling plan = 10, end-points, see 4.4.2.

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV)of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	4066	I_{FSM} = 25 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, I_O = 750 mA dc; V_R = rated V_{RWM} (see 1.3); T_A = +100°C; (T_{EC} = +100°C for surface mount devices (UL and US version)).
B3	1027	T_A = Room ambient as defined in MIL-STD-750; I_O = 1 A dc minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3, 4.5.3, and 4.5.6).
B5	3101 or 4081	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) ≤ 38°C/W, L = .375 inch (9.53 mm) (see figure 6). For surface mount device (UL and US version), $R_{\theta JEC}$ ≤ 7°C/W.
B7	---	Peak reverse power, see 4.5.5 and figure 4 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; Sampling plan = 10, end-points, see 4.4.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 and as specified in table II herein. $Z_{\theta JX}$ need not be performed. See subgroup conditions for delta limits when applicable. Leaded samples may be used in lieu of surface mount devices (UL and US version) for life tests.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Tension: Test condition A, weight = 12 pounds. Lead fatigue: Test condition E, weight = 1 pound. NOTE: Both tension and lead fatigue are not applicable for surface mount device (UL and US version).
C6	1026	T_A = +25°C minimum, I_O = 1 A dc minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3, 4.5.3, and 4.5.6). T_{EC} = +100°C to +135°C for surface mount devices (UL and US version).

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4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 (subgroups E1 thru E4 are not applicable).

Subgroup	Method	Conditions
E5	1001	1N5617, 1N5619 = 8 mm Hg (100,000 ft); 1N5621, 1N5623 = 33 mm Hg (70,000 ft); V_R = V_{RWM} (see 1.3); t = 1 minute (minimum).

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature T_A of +25°C ±3°C.

4.5.3 Life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180°, nor less than 150°.

4.5.4 Scope display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 5 to 20 µA/division and 50 to 200 V/division. Reverse current over the knee shall be at least 50 µA and less than 500 µA peak. Any discontinuity below actual breakdown or dynamic instability of the trace shall be cause for rejection of that device.

4.5.5 Peak reverse power test. This test shall be measured in the circuit of figure 7, or equivalent. A 20 microsecond half-sine waveform of current shall be used and peak reverse power shall be determined by the product of peak reverse voltage and peak reverse current.

4.5.6 Mounting conditions. At the option of the manufacturer, any clips or heat sink mounting configurations may be utilized provided that I_O is adjusted such that the junction temperature of each diode is maintained at least +120°C above the specified ambient temperature of +25°C.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1	2071					
Visual and mechanical inspection						
Subgroup 2						
Forward voltage	4011	I_F = 3 A dc (pulsed); (see 4.5.1) t_p = 300 µs; 2 percent maximum duty cycle	V_F	0.8	1.6	V_{pk}
Reverse current leakage	4016	DC method	I_{R1}			
1N5615, 1N5615UL,US		V_R = 200 V dc			0.5	µA dc
1N5617, 1N5617UL,US		V_R = 400 V dc			0.5	µA dc
1N5619, 1N5619UL,US		V_R = 600 V dc			0.5	µA dc
1N5621, 1N5621UL,US		V_R = 800 V dc			0.5	µA dc
1N5623, 1N5623UL,US		V_R = 1,000 V dc			0.5	µA dc
Breakdown voltage	4021	I_R = 50 µA dc	$V_{(BR)R}$			
1N5615, 1N5615UL,US				220		V dc
1N5617, 1N5617UL,US				440		V dc
1N5619, 1N5619UL,US				660		V dc
1N5621, 1N5621UL,US				880		V dc
1N5623, 1N5623UL,US				1100		V dc
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		4.5	°C/W
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	DC method	I_{R2}			
1N5615, 1N5615UL,US		V_R = 200 V dc			25	µA dc
1N5617, 1N5617UL,US		V_R = 400 V dc			25	µA dc
1N5619, 1N5619UL,US		V_R = 600 V dc			25	µA dc
1N5621, 1N5621UL,US		V_R = 800 V dc			25	µA dc
1N5623, 1N5623UL,US		V_R = 1,000 V dc			25	µA dc
Subgroup 4						
Reverse recovery time	4031	Condition B1	t_{rr}			
1N5615, 1N5615UL,US					150	ns
1N5617, 1N5617UL,US					150	ns
1N5619, 1N5619UL,US					250	ns
1N5621, 1N5621UL,US					300	ns
1N5623, 1N5623UL,US					500	ns
Capacitance	4001	V_R = 12 V dc, 0.1 < f < 1.0 MHz	C			
1N5615, 1N5615UL,US					45	pF
1N5617, 1N5617UL,US					35	pF
1N5619, 1N5619UL,US					25	pF
1N5621, 1N5621UL,US					20	pF
1N5623, 1N5623UL,US					15	pF
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups A, B, and C electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Reverse current leaking change	4016	DC method	ΔI_{R1} 4/		For JAN, JANTX, and JANTXV, ≤ 250 nA dc or 100 percent, whichever is greater; for JANS, ≤ 100 nA dc or 100 percent, whichever is greater.	
2	Forward voltage change	4011	I_F = 3 A dc; Pulsed (see 4.5.1) t_p = 300 µs; 2 percent maximum duty cycle.	ΔV_{F1} 4/		± 50 mV dc maximum change from previous measured value.	

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, step 2.

b. Subgroup 4, see table II herein, step 2.

c. Subgroup 5, see table II herein, step 1.

2/ The electrical measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, step 1.

b. Subgroup 6, see table II herein, step 1.

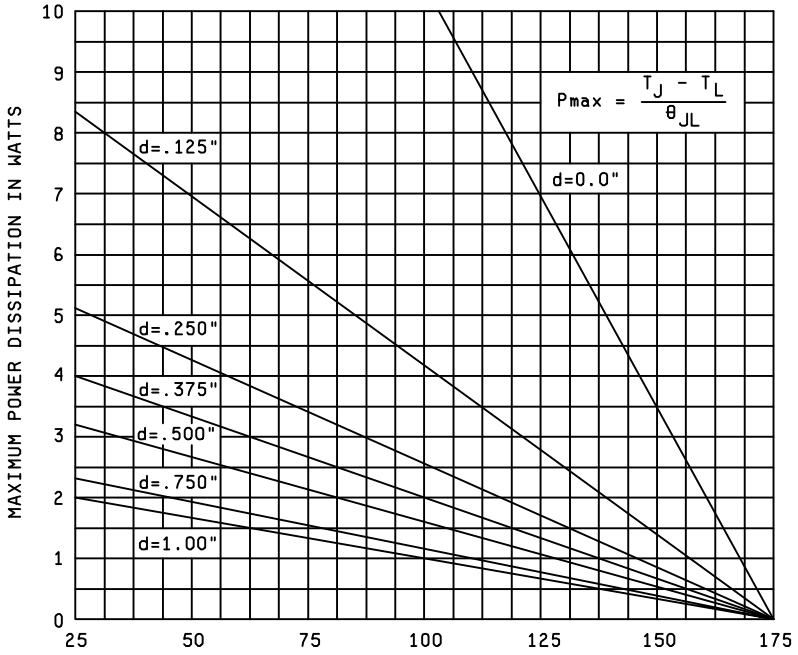
3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, step 1 (JANS).

b. Subgroup 6, see table II herein, step 1 and 2 (JANS), step 1 (JAN, JANTX, and JANTXV).

4/ Devices which exceed the group A limits for this test shall not be accepted.

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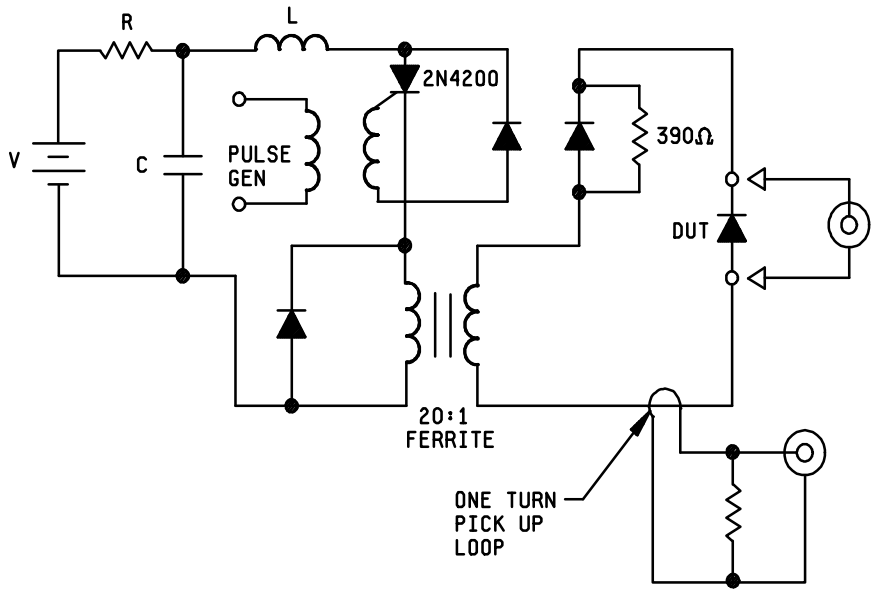
Maximum lead temperature in °C (T_L) at point d from body (see table below).
For maximum operating junction temperature of +175°C with equal two-lead conditions.

d (see notes 1 and 2)		$R_{\theta JL}$
Dimensions		°C/W
Inches	mm	
.000	(0.00)	7
.125	(3.18)	18
.250	(6.35)	29
.375	(9.53)	38
.500	(12.70)	45
.750	(19.05)	62
1.000	(25.40)	76

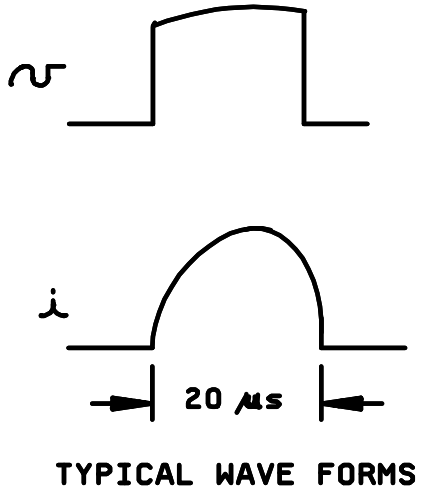
NOTES:
1. Lead distance point d and thermal resistance data.
2. Thermal resistance test performed at .375 inch (9.53 mm).
3. $R_{\theta JEC} \leq 7$°C/W.

FIGURE 6. Maximum power versus lead temperature.

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NOTES:
1. L - 13T #22 pm 1 inch (25.4 mm) diameter form (air core).
2. C - 1 to 10 µfd to give a 20 µs pulse width.
3. V - adjustable to 200 volts for power desired in D.U.T.



TYPICAL WAVE FORMS

FIGURE 7. Peak reverse power measurement circuit and waveform.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level, type designator, and for die acquisition, the JANHC and JANKC identification (see 6.3 and figures 4 and 5).

6.3 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5615) will be identified on the Qualified Manufacturers List (QML).

JANHC and JANKC ordering information		
PIN	Manufacturer	
	14552	33178
1N5615	JANHCA1N5615 JANKCA1N5615	JANHCB1N5615 JANKCB1N5615
1N5617	JANHCA1N5617 JANKCA1N5617	JANHCB1N5617 JANKCB1N5617
1N5619	JANHCA1N5619 JANKCA1N5619	JANHCB1N5619 JANKCB1N5619
1N5621	JANHCA1N5621 JANKCA1N5621	JANHCB1N5621 JANKCB1N5621
1N5623	JANHCA1N5623 JANKCA1N5623	JANHCB1N5623 JANKCB1N5623

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1825)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/429F	**2. DOCUMENT DATE** 21 April 1999

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY TYPES 1N5615, 1N5617, 1N5619, 1N5621, 1N5623, 1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, 1N5623UL, 1N5615US, 1N5617US, 1N5619US, 1N5621US, 1N5623US, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

DD Form 1426, FEB 1999 (EG) Previous editions are obsolete WHS/DIOR, Feb 99

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